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1 Logical framework

1.1 Definitions of logical notions

This subsections is devoted to construction of a simple mathematical model of a logical notions. Under construction. It follows standard definitions and the code of *checker.ml*.

In this section we deal with strings of letters. NOTE: The definitions in this section are simultaneous: some earlier definitions depend on later. It is possible to rewrite equivalently the definitions via natural number parametrised of "stages" of definitions, and then take the unions of stages to obtain the final definitions. But it is too detailed. These definitions were implemented, so they are checked for soundness by the PolyML compiler, so there is no need in such messy unwinding of the definitions.

Definition 1.1. (Variables)
The countable set
Var

Definition 1.2. (Functional symbols)
The countable set
FuncSymb

Definition 1.3. (Predicate symbols)
The countable set
PredSymb

Definition 1.4. (Term)
Inductive definition of term:
1) "v" $\in$ Var $\implies$ "v" $\in$ Term.
2) "f" $\in$ FuncSymb & "t_1" $\in$ Term & ... & "t_n" $\in$ Term $\implies$ "$f(t_1,\dots,t_n)$" $\in$ Term.
3) Some higher-order features:
"x" $\in$ Var & "φ" $\in Fm \implies$ "$(\varepsilon x:\varphi)$" $\in$
...

Definition 1.5. (Formula)
Inductive definition of formula:
1)$\bot \in$ Fm.
2)$P \in$ PredSymb & $t_1 \in$ Term & ... & $t_n \in$ Term $\implies P(t_1,\dots,t_n) \in$ Fm.
3)$A \in$ Fm $\implies (\neg A) \in$ Fm.
4)$A \in$ Fm & $B \in$ Fm $\implies (A \longrightarrow B) \in$ Fm.

5)$A \in \mathrm{Fm}$ & $B \in \mathrm{Fm} \implies (A \wedge B) \in \mathrm{Fm}$.
6)$A \in \mathrm{Fm}$ & $B \in \mathrm{Fm} \implies (A \vee B) \in \mathrm{Fm}$.
7)$x \in \mathrm{Var}$ & $A \in \mathrm{Fm} \implies (\forall x : A) \in \mathrm{Fm}$.
8)$x \in \mathrm{Var}$ & $A \in \mathrm{Fm} \implies (\exists x : A) \in \mathrm{Fm}$.

Definition 1.6. (Context)
Context = {d}

Definition 1.7. (Statement)

Definition 1.8. (Proof)
Let F be a statement. (i.e. a pair: context and formula)
Let P be a list of formulas. (i.e. function from 0,...,n-1 to Fm)

1.1.1 Substitution

...

Definition 1.9. Simultaneous substitution.

$$\varphi[Z_1/W_1, ..., Z_n/W_n]$$

1.2 Semantics

1.2.1 Motivation

The letter $\mathcal{U}$ denotes the conglomerate of all classes.

Let $\mathcal{PU} \setminus \{\emptyset\}$ be the domain of discourse on the global level of the presented model for the theory.

It means that we have a super-large model.
Such big model is simple and natural for having predicates such as (- exist), (- unique), (- defined) being defined.
Note: $(t \text{ defined}) \iff ((t \text{ exists}) \wedge (t \text{ unique}))$

Let w be an evaluation of variables:

$$w : \mathrm{Var} \to \mathcal{PU} \setminus \{\emptyset\}$$

But we are interested mostly into variable interpreted as "class-singletones" of any(even proper) class, as an actual object of interest. Here we assume existence of such conglomerates. So the our chosen naive metatheory is weakly stronger than the theory presented (in sense that it is not weaker). If we chose not to define (- exist), (- unique), (- defined), we could use the theory itself as its own metatheory. Actually, there is a translation of theorems that collapses all terms to be defined.

1.2.2 Ideas

The whole this section is devoted to definition of interpretation of terms and formulas under arbitrary assignment and under assigmnent with singletones.
We simultaneously define evaluation of formula with singleton-assignment, and extend it to arbitrary assignment:

$$\left(w \models \varphi\right) \iff \left(\forall \left(v \in \prod w\right) : (v^* \models \varphi)\right),$$

or, in different notation,

$$I^w_{\mathrm{Fm}}(\varphi) := \text{if } \left(\forall \left(v \in \prod w\right) : (v^* \models \varphi)\right) \text{ then True else False},$$

Semanntical consequence $(v^* \models \varphi)$ is defined recursively by cases of the formula φ.

$$(v^* \models (t \ \text{ unique})) \iff (I^{v^*}_{\mathrm{Term}}(t) \ \text{ semisingleton})$$

Example 1. Let $w := \{(\text{"}x\text{"}, \{0, 1\})\}$, but even in such situation, variables are keeping being defined, i.e. the following holds: $w \models (x \ \text{ unique})$.

Theorem 1.1. The definitions of interpretation coincide on singleton-assignments. For all $v : \mathrm{Var} \to \{\{q\} | q \in \mathcal{U}\}$.

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1.2.3 Lemmas for axiom validation

Some lemmas useful for proving validity of the axioms. The following lemma has several parts, which are proved simultaneously.

Lemma 1.1. (substituition equals changing the assignment)

1.2.4 Validation of axioms and rules

Theorem 1.2. (Generalization rule)

$$\frac{\models \psi}{\models (\forall x : \psi)}$$

Proof.
The premise is equivalent to the following statement:

$$\forall\, (w : \mathrm{Var} \to \mathcal{PU} \setminus \{\emptyset\}) : \forall \left(v \in \prod w\right) : \forall\, (q \in \mathcal{U}) : (v^* \models \psi)$$

Done.

Theorem 1.3. (specification rule)

$$\models ((t \text{ exists}) \wedge (\forall x : \psi)) \;\longrightarrow\; \psi[t/x]$$

Proof.
Done.

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1.2.5 OLD VERSION, NOT GOOD: Precise definitions

Definition 1.10. (Global assignment / "domain assigment")
A function w is a global assignment iff $w : \mathrm{Var} \to ((\mathcal{PU}) \setminus \{\emptyset\})$ (note, domains should be nonempty, otherwise statement will be too weak to deal with normally)

Definition 1.11. (Local assignment w.r.t. global assigment)
v is a local assignment w.r.t. global assignment w if $v : \prod_{(x:\mathrm{Var})} w(x)$,
i.e. $v : \mathrm{Var} \to \mathcal{U}$, such that $\forall (x : \mathrm{Var}) : v(x) \in w(x)$.

Definition 1.12. (Local interpretation of term under local assignment v wrt the global assignment w)
(almost classical definition)
Terms are interpreted as subconglomerates of the conglomerate of all classes.

$$I^{\mathrm{loc},v}_{\mathrm{Term}} : \mathrm{Term} \to \mathcal{U}$$

$$I^{\mathrm{loc},v}_{\mathrm{Term}}(x) := v(x)$$

$$I^{\mathrm{loc},v}_{\mathrm{Term}}(f(t_1, \ldots, t_n)) := I(f)(I^{\mathrm{loc},v}(t_1), \ldots, I^{\mathrm{loc},v}(t_n))$$

$$\ldots$$

Definition 1.13. (Local interpretation of formula) (also classical definition of "truth under assignment")

$$I_{\mathrm{Fm}}^{\mathrm{loc},v} : \mathrm{Fm} \to \{\mathrm{true}, \mathrm{false}\}$$

$$(I_{\mathrm{Fm}}^{\mathrm{loc},v}(P(t_1, \ldots, t_n)) = I(P)(I_{\mathrm{Term}}^{\mathrm{loc},v}(t_1), \ldots, I_{\mathrm{Term}}^{\mathrm{loc},v}(t_n)))$$

$$\ldots$$

Definition 1.14. (Locally valid formula w.r.t. global assignment w)
A formula is locally valid if its locally true for any local assignment v compatible with the global assignment w.

Definition 1.15. (global interpretation of terms under a global assignment w)
Remark: It is just the collection of all possible interpretations of terms wrt any local assignment that respects the global assignment.
This is kinda "useless definition", because we don't care about the collection of interpretations of terms while constructing global interpretation of formula in the next definition.

$$I_{\mathrm{Term}}^{\mathrm{glob},w}(t) := \left\{ I_{\mathrm{Term}}^{\mathrm{loc},v}(t) \middle| v \in \prod_{x \in \mathrm{Var}} w(x) \right\}$$

Definition 1.16. (Global interpretation of a formula)
A formula φ is globally true (wrt global assignment w) if its locally valid (i.e it is true w.r.t local assignments respecting the global assignment w)

$$I_{\mathrm{Fm}}^{\mathrm{glob},w} : \mathrm{Fm} \to \{\mathrm{True}, \mathrm{False}\}$$

$$\left(I_{\mathrm{Fm}}^{\mathrm{glob},w}(\varphi) = \mathrm{True}\right) \iff \left(\forall \left(v \in \prod_{x:\mathrm{Var}} w(x)\right) : I_{\mathrm{Fm}}^{\mathrm{loc},v}(\varphi) = \mathrm{true}\right)$$

Example 1.1. (global interpretation of some special formulas under global assignment)

$$I_{\mathrm{Fm}}^{\mathrm{glob},w}((t \text{ exists})) := (\text{if } (I_{\mathrm{Term}}^{\mathrm{glob},w}(t) \text{ nonempty}) \text{ then True else False})$$

$$I_{\mathrm{Fm}}^{\mathrm{glob},w}((t \text{ unique})) := (\text{if } (I_{\mathrm{Term}}^{\mathrm{glob},w}(t) \text{ semisingleton}) \text{ then true else false})$$

Definition 1.17. (Strictly globally valid formula)
A formula φ is (strictly) globally valid if its globally true for any global assignment w.

$$I_{\mathrm{Fm}}^{\mathrm{strictlyglob}} : \mathrm{Fm} \to \{\mathrm{strictlyValid}, \mathrm{strictlyInvalid}\}$$

$$\left(I_{\mathrm{Fm}}^{\mathrm{strictlyglob}}(\varphi) = \mathrm{strictlyValid}\right) \iff \left(\forall (w : \mathrm{Var} \to ((\mathcal{PU}) \setminus \{\emptyset\})) : I_{\mathrm{Fm}}^{\mathrm{glob},w}(\varphi) = \mathrm{True}\right)$$

Definition 1.18. (Weakly globally valid formula)
A formula φ is globally valid if it's globally true for all such global assignment w, which satisfy that range of w consists only of conglomerate-singletones.
I.e. a formula φ is called globally valid, iff it is true under any global assignment by singletones $w : \mathrm{Var} \to \{\{K\} | K \in \mathcal{U}\}$.

$$I_{\mathrm{Fm}}^{\mathrm{glob}} : \mathrm{Fm} \to \{\mathrm{Valid}, \mathrm{Invalid}\}$$

$$\left(I_{\mathrm{Fm}}^{\mathrm{glob}}(\varphi) = \mathrm{Valid}\right) \iff \left(\forall (w : \mathrm{Var} \to \{\{K\} | K \in \mathcal{U}\}) : I_{\mathrm{Fm}}^{\mathrm{glob},w}(\varphi) = \mathrm{True}\right)$$

Using singletone-valid formulas we avoid the need of explicitly state that every variable is defined.

1.2.6 Application for "greek"–operators

We have not always well-defined epsilon operator.
(tau operator is always well-defined, epsilon operator is well-defined only on bounded formulas which specify "singletone-conglomerates"(if we could collect classes).

Here we come to the point, where we nee more relaxed notion of truthness, to include $P((\varepsilon X : \varphi))$, when predicate P is true for all classes "X" satisfying φ.
It is interesting, that quasi-equality is naturally arises as... oh no! We need to introduce a mix "exists and forall" quantifier and semantics, so the quasi-equality will natuarally arise. I mean the following:

Definition 1.19. (Interpretation of an atomic formula under some fixed global assignment) Fix some global assignment function $w : \mathrm{Var} \to \mathcal{PU}$.

- $I^w_{\mathrm{Fm}}(P(t_1, ..., t_n)) :=$ True, iff two conditions are satisfied:
  1) $I^w_{\mathrm{Term}}(t_j)$ is non-empty
  2) for all sequences of object $(q)^n_{j=1}$, such that for all j from 1 to n, $q_j \in I_{\mathrm{Term}}(t_j)$: $I_{\mathrm{PredSymb}}(P)(q_1, ..., q_n) =$ True.
- $I^w_{\mathrm{Fm}}(P(t_1, ..., t_n)) =$ False, iff two conditions are satisfied:
  1) $I_{\mathrm{Term}}(t_j)$ is non-empty
  2) for all sequences of object $(q)^n_{j=1}$, such that for all j from 1 to n, $q_j \in I_{\mathrm{Term}}(t_j)$: $I_{\mathrm{PredSymb}}(P)(q_1, ..., q_n) =$ False.
- $I^w_{\mathrm{Fm}}(P(t_1, ..., t_n)) =$ Uncertain, in all other cases.
  For example, if there exists $j \in \{1, ..., n\}$, such that $I_{\mathrm{Term}}(t_j)$ is empty;
  or if neither of the second requirements hold,
  i.e. if the function $\big((q)^n_{j=1} \mapsto I_{\mathrm{PredSymb}}(P)(q_1, ..., q_n)\big)$ is not a constant on $I_{\mathrm{Term}}(t_1) \times \cdots \times I_{\mathrm{Term}}(t_n)$.

Assume a, b are well-defined sets.
Then $(\tau x : x \in \{a, b\})$ is well-defined, since using it assumes existence of a choice conglomerate-function.
Meanwhile, $(\varepsilon x : x \in \{a, b\})$ is undefined. But it is not (semantically) equal to undef.
It is neither equal, nor not equal in theory, because, some(in fact both) sides are not well-defined.
But they are quasi-equal (syntactically), since they are both not well-defined.

Remark: Here by semantical equality i mean the equaly of all interpretations of two term under any assignment of values.
Assigment maps each variable to some singletone.
Let's come back to: $(f$ family$) \wedge (x \in \mathrm{dom}(f)) \longrightarrow (f(x) = f(x))$
For the case when something goes wrong, "(f(x)=f(x))" is not true, since it being true assumes both-hand-sides to be defined.
In order for lemmas from (link shen-logic-part2-2.pdf , page 137) to work, we need assign non-sigleton sets to variables.
But we want to avoid tedious assumtions like the following: $(x_1$ defined$) \wedge ... \wedge (x_n$ defined$) \longrightarrow \varphi$.

Therefore for theorems on the upper level assigment are considered to be with singletones only, but within the evaluation variables may be assigned to a set with non-unique values.
The previous comment about introducing constants will patch the need for shortening of the statement while it is not possible to do via variable.
But what about the cases when it should be a term, depending on some parameters?
Resolution: we use only terms, and this mechanism of definitions of terms is outside the logic of the system.
Or we can introduce $:\equiv$ somehow? YES.

Definition 1.20. (interpretation of definitional equality)
~~$I^v_{\mathrm{Fm}}(A \equiv B) := (I^v_{\mathrm{Term}}(A) = I^v_{\mathrm{Term}}(B))$~~.NO! WRONG!
$I^v_{\mathrm{Fm}}(A \equiv B) := \forall(w \text{ evaluation}) : (I^w_{\mathrm{Term}}(A) = I^w_{\mathrm{Term}}(B))$.
Will such definition work with a variable? I highly doubt. So it's better to keep such definitions of undefined values outside of the logic until there will be an explicit necessity for them. (Probably there will be no.) But this definition works well for defining functional symbols, and, in particular, constants.

Some statements:
1) VALID, TRUE: $(\varepsilon x : x \in \{a, b\}) \in \{a, b\}$.
2) INVALID: $(\varepsilon x : x \in \emptyset) \in \emptyset$. It is neither true, nor false, because the interpretation of the first term is an empty conglomerate of classes.
3) INVALID: $(\varepsilon x : x \in \emptyset) \notin \emptyset$. It is neither true, nor false, because the interpretation of the first term is an empty conglomerate of classes.
4) VALID, TRUE: $(x \in y) \longrightarrow (x \in y)$.

1.3 Axioms for the corresponding prediacate calculus

1.4 Propositional part

For all atomic formulas A, B, C.

$$(A \longrightarrow (B \longrightarrow A))$$

$$((A \longrightarrow (B \longrightarrow C)) \longrightarrow ((A \longrightarrow B) \longrightarrow (A \longrightarrow B)))$$

$$((A \wedge B) \longrightarrow A)$$

$$((A \wedge B) \longrightarrow B)$$

$$(A \longrightarrow (B \longrightarrow (A \wedge B)))$$

$$(A \longrightarrow (A \vee B))$$

$$(B \longrightarrow (A \vee B))$$

$$((A \longrightarrow C) \longrightarrow ((B \longrightarrow C) \longrightarrow ((A \vee B) \longrightarrow C)))$$

$$(\neg A \longrightarrow (A \longrightarrow B))$$

$$((A \longrightarrow B) \longrightarrow ((A \longrightarrow \neg B) \longrightarrow \neg A))$$

1.5 Predicational part

Specification rules:

$$(t \text{ defined}) \longrightarrow ((\forall x : \varphi) \longrightarrow \varphi[t/x])$$

$$(t \text{ defined}) \longrightarrow (\varphi[t/x] \longrightarrow (\exists x : \varphi))$$

Bernays' rules: if x is not free in formula ψ then

$$\frac{\psi \longrightarrow \varphi}{\psi \longrightarrow (\forall x : \varphi)}(\text{B1})$$

$$\frac{\varphi \longrightarrow \psi}{(\exists x : \varphi) \longrightarrow \psi}(\text{B2})$$

2 Useful metatheorems

Theorem 2.1. (Deduction theorem)

$$(A \vdash B) \Longrightarrow (\vdash A \to B)$$

Proof.
This theorem is provable constructively. This is done through the formalisation of a proof notion. Proof is ... **Done.**

Theorem 2.2.

$$A = B \Longrightarrow (t[A/W] = t[B/W])$$

Proof.
Assume $A = B$. By induction on term construction.
Case 1(Base): $t = v \in \text{Var}$
If also the variable v is W then $t[A/W] = A$ and $t[B/W] = B$.
Since $A = B$, $t[A/W] = t[B/W]$, so we are done. Otherwise, i.e. if the variable v is not W, then $t[A/W] = v = t$ and $t[B/W] = v = t$.
Therefore $t[A/W] = t[B/W]$, so we are done.
Case 2(Step of induction): $t = f(t_1, ..., t_n) \in \text{Term}$, where f is a functional symbol of arity n.
Now we can use induction hypotheses: $t_1[A/W] = t_1[B/W], ..., t_n[A/W] = t_n[B/W]$ and definition of substitution.

$$t[A/W] = f(t_1, ..., t_n)[A/W] =$$

$$= f(t_1[A/W], ..., t_n[A/W]) =$$

$$= f(t_1[B/W], ..., t_n[B/W]) =$$

$$= f(t_1, ..., t_n)[B/W] = t[B/W]$$

Therefore $t[A/W] = t[B/W]$, so we are done.
Done.

Theorem 2.3.

$$A = B \longrightarrow (\varphi[A/W] \Longleftrightarrow \varphi[B/W])$$

Proof.
By induction on formula construction.
...
Done.

2.1 Extension by definitions

...

2.2 Soundness of the calculus

...

3 First-order logic[skip this section; go to sect.3]

...

Theorem 3.1. In intuitionistic logic:
$\neg(A \vee B) \longrightarrow (\neg A \wedge \neg B)$

Proof.
Using deduction theorem it is enought to show (backward reasoning):
$\neg(A \vee B) \vdash (\neg A \wedge \neg B)$ (by OR-elim).
Subgoal 1: $\neg(A \vee B) \vdash \neg A$ (by deduction)
$\neg(A \vee B), A \vdash \bot$ (by MP $(A \vee B) \longrightarrow \bot$)
$\neg(A \vee B), A \vdash (A \vee B)$ (by OR-intro)
$\neg(A \vee B), A \vdash A$ (by assumption).
Subgoal 2: $\neg(A \vee B) \vdash \neg B$ (by deduction).
$\neg(A \vee B), B \vdash \bot$ (by MP $(A \vee B) \longrightarrow \bot$).
$\neg(A \vee B), B \vdash (A \vee B)$(OR-intro).
$\neg(A \vee B), B \vdash B$ (by assumption).
Done.

Theorem 3.2. In intuitionistic logic:
$\neg\neg(P \vee \neg P)$

Proof.
Using deduction theorem it is enought to show (backward reasoning):
$\neg(P \vee \neg P) \vdash \bot$ (cut rule with **??**)
$\neg P \wedge \neg\neg P \vdash \bot$ (by Left_AND_Intro)
$\neg P, \neg\neg P \vdash \bot$ (by MP: $\neg P \longrightarrow \bot$).
$\neg P, \neg\neg P \vdash \neg P$ (by assumption).
Done.

Theorem 3.3.

$$(\exists x \forall y : \varphi) \longrightarrow (\forall y \exists x : \varphi)$$

Proof.
Assume $(\exists x \forall y : \varphi)$. To show: $(\forall y \exists x : \varphi)$. (deduction theorem is in use)

$$(\exists x \forall y : \varphi) \implies (\forall y : \varphi) \quad (\exists \text{E, (x/x)})$$

Fix y. To show: $(\exists x : \varphi)$. (or just do $\forall I$ in the end)

$$(\forall y : \varphi) \implies \varphi \quad (\forall \text{E, (y:=y)})$$
$$\varphi \implies (\exists x : \varphi) \quad (\exists \text{I, (x:=x)})$$

Done.